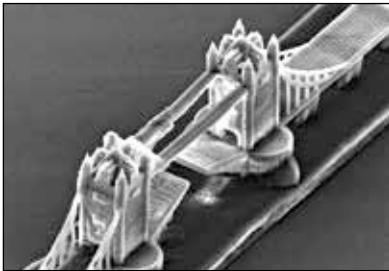


tion usually involve a great deal of consideration in relation to the how of manufacturing; questions such as do we have the right tools and do we have the right machines become no longer an issue with 3D printing. The imagination to design and create is limitless and negligible in costs. In traditional setups a furniture designer would be limited by the machines available such as a milling machine or a sander, and would have to design based on those restrictions, however with 3D printers they can design with full freedom and allow exploration and creation of new unlikely designs. Using this freedom we already see shapes and designs only found in nature becoming a part of the designers inspiration.

Potential 6: Traditional manufacturing requires a large number of people with an array of skillsets to make any product a reality; the designers themselves would need years of hands on training with materials before they have the necessary skills to replicate even a basic object. With the use of computer guided machines and computer aided design software this limitation is eased since the design files are all that a 3D printer needs to produce any product. This freedom to create with unskilled manufacturing allows products to be accessible in areas where skill is limited or remotely placed.

Potential 7: Traditional manufacturing calculates productivity based on the quantity of creation per volume of production space. In most cases this



Even on a nano-scale this model of London's Tower Bridge isn't beyond reality. Made by Vienna Institute of Technology.

means that large manufacturing machines can only create products or parts that are considerably smaller than the machines themselves. But in the world of 3D printing this barrier is no longer true. Even a small 3D printer has greater productivity per volume of production space and can be arranged in a configuration of autonomously mobile printing apparatus so that they can print objects much larger

than the printer themselves. This form of high production capacity allows 3D printers to find utility in a home as well as a business.

Potential 8: As we've already discussed, the vastness of traditional manufacturing leads to many other costs, a significant part of which are by-product waste material. In the case of 3D printing, the precision and method